

Yutong Wu

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EDUCATION

- **Rice University** August 2023 - Present
Computer Science Ph.D. Houston, TX
 - GPA: 3.94/4.00
 - Research on agent alignment and Computational Social Science, studying how to align agent behavior with human values and sustain cooperation within human societies, among AI agents, and in human-agent interactions
- **Rochester Institute of Technology** August 2019 - May 2023
Computational Mathematics, B.S. Rochester, NY
 - GPA: 3.88/4.00
 - Minor: Computer Science

RESEARCH EXPERIENCE

- **Reputation and the Emergence of Cooperation in Sequential Social Dilemmas (SSD)** May 2025 - Present
Agent Alignment, Reinforcement Learning, Opinion Dynamics, Game Theory 
 - Designed a reputation-based alignment framework that leverages implicit rewards to steer agent behavior toward cooperation without hand-specified objectives
 - Modeled multiple reputation-broadcasting protocols by integrating the DeGroot opinion dynamics model to study how social signals shape alignment
 - Reimplemented the SSD games as an open-source project, resolving compatibility issues in the deprecated version
 - Built the multi-agent environment with PettingZoo and trained DQN agents in PyTorch
- **Engagement, Collaboration, and Conflict in Three Editions of Reddit r/place** May 2023 - May 2025
Data Analysis, Clustering, Social Networks 
 - Under review at CSCW
 - Conducted a cross-edition quantitative analysis across three iterations of the r/place experiments
 - Designed a scalable graph-based clustering framework to uncover failed coalitions using Graph-Based Image Segmentation (GBIS), Node2Vec, and Set Cover
 - Analyzed participation patterns, coordination inefficiencies, and adversarial interactions at scale

SERVICE & TEACHING

- **Undergraduate Research Mentor** January 2025 - May 2025
Rice University Houston, TX
 - Supervised an undergraduate project analyzing Reddit communication data, extending the r/place study
 - Led weekly research meetings and mentored the student in time series analysis and network science
- **Teaching Assistant** August 2024 - May 2025
Rice University Houston, TX
 - TA for COMP 559 - Graph Machine Learning (Spring '25) and COMP 614 - Programming for Data Science (Fall '24)
 - Led weekly discussion sections, held office hours, and graded assignments

WORK EXPERIENCE

- **Justice & Security Strategies** [🌐]

September 2022 - November 2022

Consultant

Remote

- Worked as a part-time contractor to analyze the turnout time of the vehicles of the Los Angeles Fire Department
- Took initiative in the project and performed data analysis using Python and Jupyter Notebook
- Produced evaluation results from the training data and compared the results with the testing data

- **Anchorage Digital** [🌐]

May 2022 - August 2022

Software Engineering Intern

Remote

- Worked in the Protocols team as a software engineering intern on implementing crypto assets into the ecosystem
- Scaffolded a tender mint-like blockchain using Ignite CLI that allows users to fund, transact, and check balances
- Configured and spun up the blockchain inside a Docker container using docker file and shell script
- Built features using Go and Protobuf to connect the blockchain and the services of the Anchorage ecosystem

- **Tencent** [🌐]

May 2021 - August 2021

Back-end Developer Intern

Remote

- Worked at the Nature Base(NBase) department on a communication app centered on group chats, mainly using Go
- Worked closely with the Big Data team to help better analyze user behaviors
- Improved the security of the app to prevent web crawlers from fetching private user data
- Built a whitelist feature for special experiences to certain users by working closely with the front-end team